

MASSACHUSETTS INSTITUTE OF TECHNOLOGY
DEPARTMENT OF MECHANICAL ENGINEERING
2.671 Measurement and Instrumentation

Uncertainty Analysis Part 1: Basics and Parameter Uncertainty

Many of the learning objectives in 2.671 relate to how to analyze and present experimental uncertainty. A series of three Uncertainty Analysis documents introduce you to basic concepts in uncertainty analysis and then shows you how to apply them to estimate uncertainty on parameters derived from measurements. The material in this document will be used to complete the Soda Can Postlab Part 1. Uncertainty Analysis Part 2 describes how to fit model functions to your data, including linear, polynomial, exponential, power law, etc. Topics that will most likely be essential for your Go Forth and Measure data analysis, such as model selection, determination of uncertainty in fit parameters, and uncertainty in values predicted from a fit model are covered in Uncertainty Analysis Part 2. Uncertainty Analysis Part 3 describes how to compare different populations and evaluate whether they are statistically the same or different, and, to quantify the difference. You will need to read Part 3 before doing the Soda Can Postlab Part 3.

The uncertainty analysis documents will be indispensable to you in the analysis of your Go Forth and Measure results, and the concepts will be further reinforced in future labs. This document can be found online in the Uncertainty page of the Wiki as a PDF file. We will give you pointers on how to perform calculations with Logger Pro, Excel, and MATLAB.

Essential information in each document has been extracted into a Slide Deck that should make it easier for you to perform uncertainty analysis. Please refer to the longer text documents for further details and references.

Please also refer to the Uncertainty Analysis page of the Wiki, where you can more easily determine which uncertainty analysis method is suitable for your given problem by reading the descriptions of the four CASES listed there.

1. Basic Concepts of Statistics [1]

For much of your 2.671 work, you will record multiple measurements of the same quantity. The mean (average) of your measurements is typically reported as the *mean measured value*. The uncertainty in the mean measured value can be related to the standard deviation of the measurements, as described in Sect. 4. In this section, we remind you of the definitions of mean and standard deviation.

Assume that n measurements have been taken of a variable x . The mean measured value μ_x is given by

$$\mu_x = \sum_{i=0}^{n-1} \frac{x_i}{n}, \quad (1)$$

where x_i is the result of the i^{th} measurement. Note that this is the mean value of the sample set, not of the true population of possible measured values (assumed to be infinite in extent and thus not measurable). The *sample variance* ν_x is given as

$$\nu_x = \sum_{i=0}^{n-1} \frac{(x_i - \mu_x)^2}{n-1}, \quad (2)$$

and the *sample standard deviation* σ_x is given by the square root of the variance as

$$\sigma_x \equiv \sqrt{v_x} = \sqrt{\sum_{i=0}^{n-1} \frac{(x_i - \mu_x)^2}{n-1}}. \quad (3)$$

It is also possible to calculate the standard deviation of the population for a large number of measurements. In this case, the factor of $n - 1$ in the denominator of Eqs. (2) and (3) is replaced by n . It is apparent that for large numbers of measurements, the sample and population standard deviations are essentially identical, but they can be rather different for small numbers of samples.

In 2.671, you will typically perform a relatively small number of measurements. Therefore, you should use the **sample standard deviation** defined in Eq. (3) for all uncertainty calculations. Logger Pro allows you to graphically determine statistical properties of your data, or a subset of your data, using the Statistics button on the Toolbar. Excel includes the functions **STDEV.S** (equivalent to **STDEV**) and **STDEV.P** for calculation of the sample and population standard deviations, respectively. In MATLAB, the corresponding commands are **std(x)** [equivalent to **std(x,0)**] and **std(x,1)**. Make sure to use the sample standard deviation, **std(x)** for your 2.671 work. If you wish to determine standard deviation with your calculator, please verify that the result is the **sample standard deviation**, Eq. (3).

2 Probability Density Function and the Gaussian (Normal) Distribution

The mean and standard deviation of a set of measurements, while important, do not completely describe the data. Fig. 1 shows two data sets, x and y , each of which has 1000 values with identical mean and standard deviation. We will refer to each data set as a “signal”.

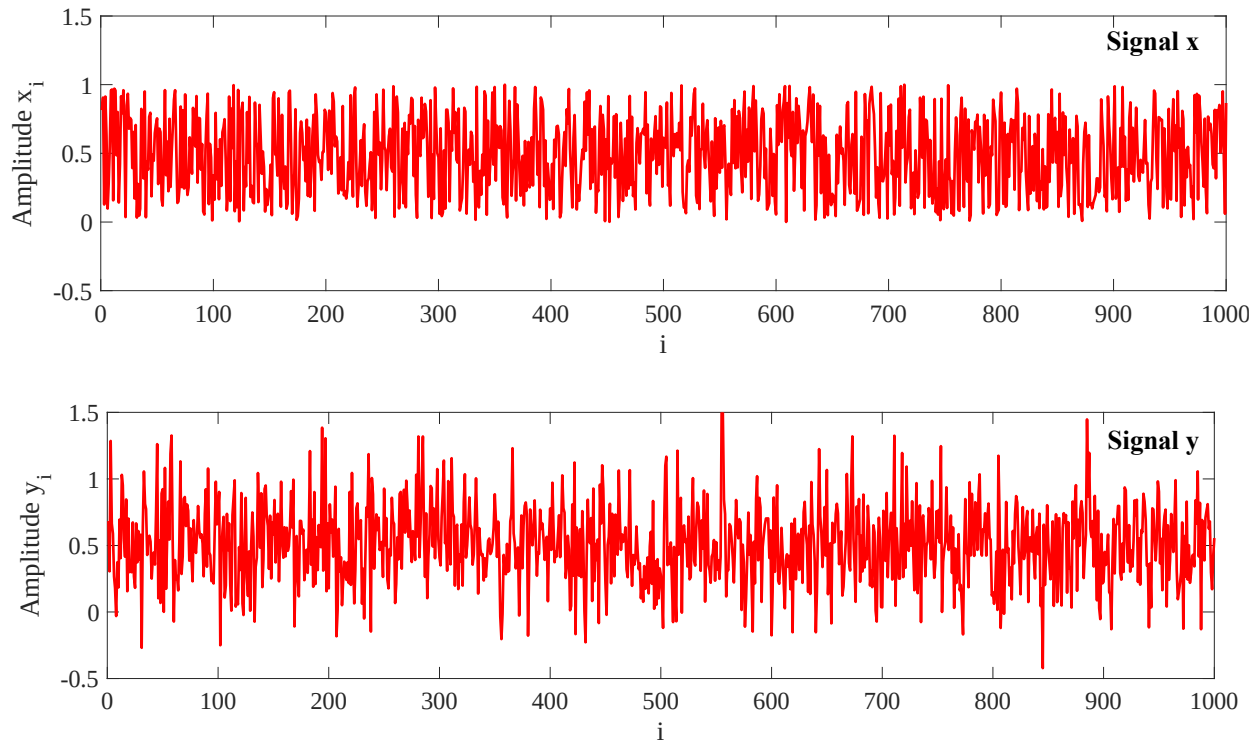


Figure 1: Two signals, x and y that each comprise 1000 values and have identical mean and standard deviation, but are obviously different signals.

These signals are obviously different, even though they have identical mean and standard deviation. In order to gain more information about each signal, we define the **probability density function**, shown in Fig. 2, which is a normalized measure of the distribution of amplitude values in the signal. The independent variable (horizontal axis) represents the amplitude of the signal, and the dependent variable (vertical axis) represents the probability density (defined as the probability per unit amplitude) of a given amplitude.

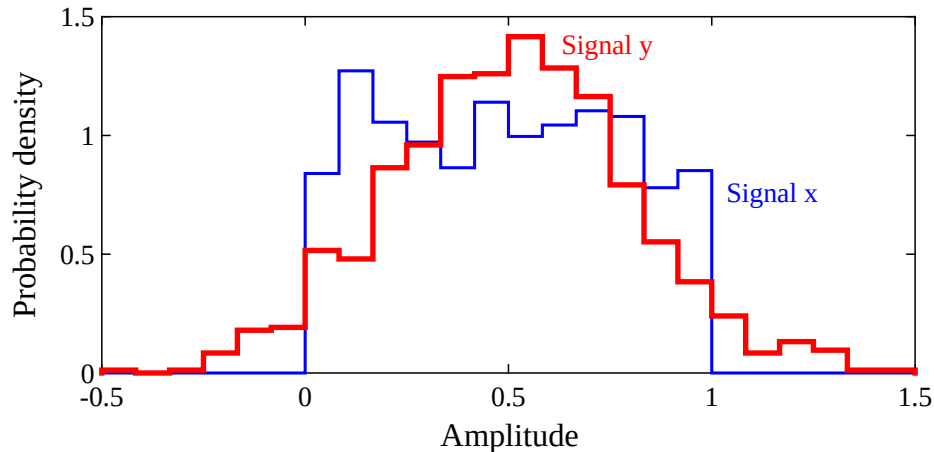


Figure 2: Amplitude probability density of the two signals shown in Fig. 1

First of all, the integral of the probability density over all values of amplitude (from $-\infty$ to $+\infty$) is, by definition, 1, or 100%. This can be determined by inspection for signal x in Fig. 2, which can be approximated as a rectangle with a constant probability density of 1 for amplitudes between 0 and 1. Comparison of the area under the curve for signal y to that for signal x shows that this relationship holds true for signal y as well. Signal y in fact has a common probability density function, called the Gaussian probability density function. The Gaussian function is defined in Eq. (4) where μ is the mean value of the signal and σ is the standard deviation. The Gaussian function with the same μ and σ as signal y and the probability distribution for signal y are shown in Fig. 3. It is apparent that signal y is essentially Gaussian.

$$P_{\text{Gauss}}(x, \mu, \sigma) \equiv \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad (4)$$

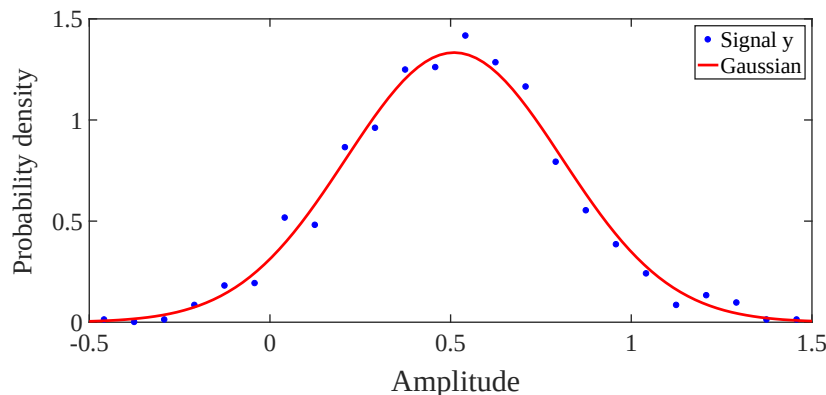


Figure 3: Comparison of probability density function for signal y and a Gaussian function with the same mean and standard deviation ($\mu = 0.5$, $\sigma = 0.3$)

The second important property of the probability density function is that the probability that the amplitude is between two values, x_1 and x_2 is given by the integral of the probability density function between x_1 and x_2 . We are usually interested in the probability that the amplitude is within some range $\pm\delta$ of the mean value, i.e. between $\mu-\delta$ and $\mu+\delta$. We will refer to this quantity as the “confidence integral”, C , for reasons that will become apparent later. The confidence integral for a Gaussian distribution is defined in Eq. (5).

$$C_{Gauss}(\delta, \mu, \sigma) \equiv \int_{\mu-\delta}^{\mu+\delta} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx \quad (5)$$

The range parameter, δ , is typically expressed in terms of the standard deviation, σ , to allow comparison between Gaussian functions with different values of σ . The confidence integral for $\delta = \sigma$, given by the integral of a region $\pm 1\sigma$ around the mean, is found from Eq. (5) as 68%. This region is shaded blue in Fig. 4. Therefore, the probability that an additional measurement of the amplitude of signal y falls within $\pm 1\sigma$ of the mean μ of all the measurements represented by the data set shown in Fig. 1-3 is 68%. Stated another way, one can state “**with 68% confidence**” that a measurement of the amplitude of signal y will be within 1σ of the previously measured mean value, or, 68% of the time such a measurement is made, the result will be within one standard deviation of the mean value.

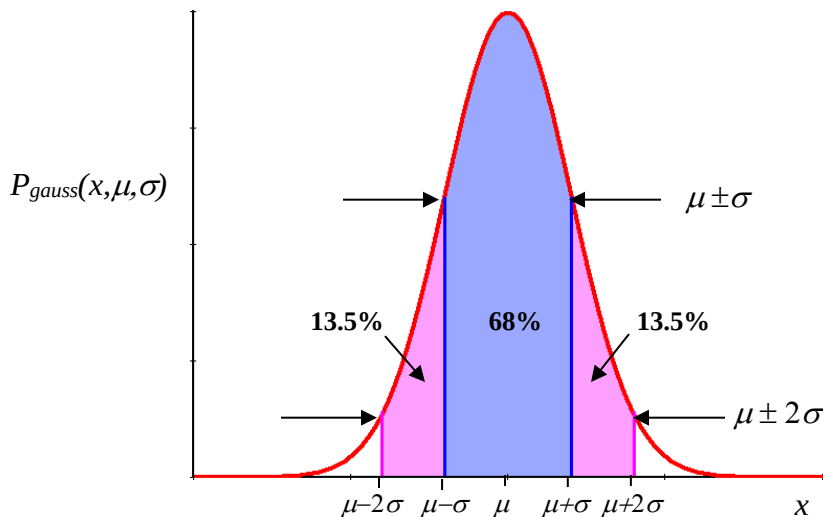


Figure 4: Gaussian probability distribution with mean μ and standard deviation σ . The blue shaded area indicates that 68% of the values fall within $\pm 1\sigma$ of the mean, and the combination of the blue and pink shaded areas indicates that 95% of the values fall within $\pm 2\sigma$ of the mean.

Table 1 gives the value of the confidence integral for several values of δ , in terms of the standard deviation, σ . One can tell from Table 1 that 99% of the samples are included in the range $\mu \pm 2.6\sigma$. In 2.671 we use 95% confidence intervals, which corresponds to a range around the mean of approximately $\pm 2\sigma$, so you may see this referred to as the “2-sigma” uncertainty.

Table 1: Value of Gaussian confidence integral defined in Eq. (5). In 2.671 we will typically use the 95% confidence interval, approximated as $\delta = 2\sigma$.

distance from mean δ	Confidence integral C_{Gauss}
1σ	68%
$1.96\sigma (\cong 2\sigma)$	95%
2.6σ	99%
3.3σ	99.9%
3.90σ	99.99%

3. Precision, Accuracy, and Discrepancy

Before we teach you how to compute your experimental uncertainty, you must realize that there are two different measures of the "goodness" of your data: **precision** and **accuracy**.

Precision refers to the repeatability of the results. If you perform the same experiment under the same conditions a large number of times, the precision can be determined from the spread in the results. In other words, random errors in the data decrease the precision of the measurement. The precision uncertainty is related to the standard deviation of the data, as shown in Sect. 4 below, and can be computed from a statistical analysis of your measurements.

Accuracy is defined as the difference between the measurement and the "true" value. In many cases, this is difficult to determine, because the true value is not known (why would you measure something you already know?) Systematic errors result in decreased accuracy. Systematic errors are much more difficult to identify than random errors, since they introduce a constant shift in the results, rather than a scatter in the results. Without a priori knowledge of the "true" result, a shift in the measured value is essentially impossible to detect. In practice, accuracy of the measurement system is usually determined using standards. A standard is a sample to be measured for which the measured quantity is known with high precision and accuracy, either through measurement with a different technique, or by a well-benchmarked and widely accepted analytical calculation. Figure 5 presents a graphical description of the difference between precision and accuracy.

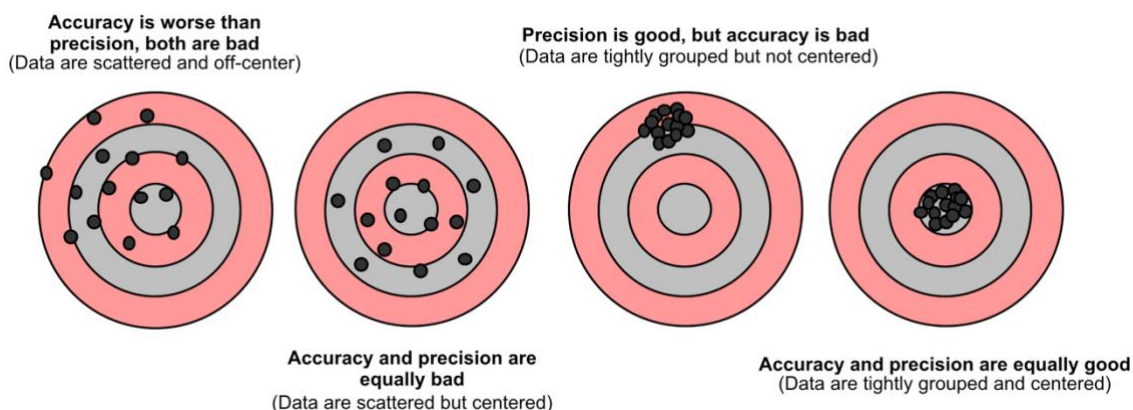


Figure 5: Graphical representation of the difference between precision and accuracy.

In many of the 2.671 experiments you will be comparing your data to "expected" values obtained from the literature (i.e. the thermal expansion coefficient of aluminum) or calculated from theory (i.e. the speed of sound in an ideal gas). These expected values contain uncertainties of their

own (i.e. is your wire sample of the same composition and treatment as that used in the literature? is the ideal gas approximation valid for your measurements?). Therefore, it is not reasonable to assume that the expected value is the “true” value for your particular measurement, and thus the difference between measured and expected values should not be referred to as the accuracy of the measurement. Rather, you should refer this difference, defined as **measured** – **expected**, as the measurement **discrepancy**. Notice that the sign of the discrepancy is important, so you must always define discrepancy as stated. Determination of the **discrepancy** is straightforward once you obtain the expected value for the measured quantity. Determination of **precision** from your measurements is more complicated and is described in the next section.

4. Computing Precision Uncertainty

There are two general types of measurements for which you will need to compute experimental uncertainty. In the first type, a quantity of interest (x) is measured multiple times. The average of the measurements μ is used to give a value for the unknown quantity. In this case, the experimental uncertainty is not the precision uncertainty in x , as discussed in Sect. 2, but the precision uncertainty in the mean of x . These uncertainties are not the same, and, in fact, the standard deviation in the mean can be significantly smaller than that of the individual measurements, and thus the corresponding uncertainty is also smaller. This approach will be referred to as the “**statistical**” method and is described in Sect. 4.1.

In the other type of experiments, the quantity of interest can only be measured once. This is true for destructive measurements, such as measuring the pressure inside a soda can by opening the can and measuring a voltage that is proportional to the change in strain. In this case, if the desired quantity depends on a number of measured variables (can diameter, can thickness, and relationship between measured voltage and strain), and if the precision of each component variable is known, either from instrument specifications or repeated measurements, the precision uncertainty of the final result can be found mathematically by propagating the uncertainty of each of the component values. This approach will be referred to as the “**propagation of uncertainty**” method and is discussed in Sect. 4.3.

An extension of the **statistical method** occurs when a quantity of interest, y , is measured as a function of another parameter, x . Fitting techniques can be used to determine the functional dependence of y on x and the uncertainty of the best-fit parameters. Often the quantity of interest is one or more of the best fit parameters, rather than the average value of y or x . The parameter uncertainty can be used to determine the **goodness of fit**. This approach is discussed in Sect. 5.

4.1 Statistical Method (CASE 1 on the Wiki Uncertainty Analysis page)

In the **statistical method**, the quantity of interest is measured multiple times and the mean and standard deviation are determined from the measurements. Calculation of the uncertainty from the standard deviation depends on the nature of the probability density function for the underlying signal. Fortunately, as will be discussed in more detail in lecture, the probability density function for most physical processes is Gaussian. (One notable exception is the distribution of the height of ocean waves, which has a higher probability for amplitudes many standard deviations greater than the mean than does the Gaussian distribution. The consequences of this can be unfortunate for those fishing on a rocky shoreline who unexpectedly get swept into the ocean because they expected a Gaussian distribution.) For the remainder of this document we will assume a Gaussian probability density, but please keep this fundamental assumption in mind and realize that the approach presented below is only valid for a Gaussian distribution.

If a set of measurements is repeated, the new average will differ somewhat from the previous one. A consequence of the *weak law of large numbers*, is that for a large number of measurements with standard deviation σ , the standard deviation S_μ of the **mean of the samples** μ approaches

$$S_\mu = \frac{\sigma}{\sqrt{n}}, \quad (6)$$

where n is the number of measurements and σ is the standard deviation of the measurements. [2]

Therefore, the standard deviation of the mean is smaller than the standard deviation of the individual measurements. For a large number of measurements, the standard deviation of the mean is **significantly smaller** than the standard deviation of the individual measurements. This is a critical result that mathematically explains the intuitive feeling that multiple measurements will provide increased experimental precision.

We described in Sect. 2 how the standard deviation relates to the confidence with which one can predict that a measurement will fall within some range of the mean of previous measurements, i.e. we have 95% confidence that a new measurement will be within $\pm 2\sigma$ of the mean. The same approach is used to describe the confidence with which the mean of a particular measurement set corresponds to the “true” mean of the measurement (i.e. the mean for an infinite number of samples). In other words, we can state with 95% confidence that the “true” mean, μ_{true} is within 2 standard deviations, S_μ of the measured mean value, μ_{meas} , or

$$\begin{aligned} \mu_{true} &= \mu_{meas} \pm 2S_\mu \\ &= \mu_{meas} \pm 2 \frac{\sigma}{\sqrt{n}}. \end{aligned} \quad (95\% \text{ confidence}) \quad (7)$$

This equation can be generalized for arbitrary confidence intervals as

$$\mu_{true} = \mu_{meas} \pm u_\mu, \quad (8)$$

where u_μ is the “precision uncertainty” for the stated confidence level derived from Eq. (6) and Table 1. The precision uncertainty is listed in terms of the standard deviation σ of n measurements in Table 2 for several levels of confidence.

Table 2: Value of precision uncertainty of the mean, u_μ , as a function of confidence level and standard deviation of the measurements σ

u_μ	with ____ % confidence
$\sigma/\sqrt{n}$	68%
$1.65\sigma/\sqrt{n}$	90%
$2\sigma/\sqrt{n}$	95%
$2.6\sigma/\sqrt{n}$	99%

It is important to realize that the stated uncertainty only considers statistical (or random) fluctuations in the measurements and thus gives the precision uncertainty. This approach cannot be used to provide any estimate of systematic errors, or accuracy.

Recall that the above analysis is only valid for Gaussian probability distributions. Fortunately, the *central limit theorem* [1], an important theorem in statistics, states that for a large number of samples, the sum of samples for any arbitrary distribution approaches that of a Gaussian. This

implies that, for samples drawn from an arbitrary distribution, the mean of those samples will be normally distributed, even if the underlying data is not normally distributed (i.e. has a Gaussian probability distribution). Recall that the sample mean is merely the sum of samples divided by the number of samples.

Therefore, the values for u_μ given in Table 2 can be used even for non-Gaussian distribution. We will typically use the 95% confidence interval in 2.671, defined by Eq. (7). Further discussion of the confidence interval for large values of n can be found in Sect. 3.6.2 of Ref. 1. The difference between the uncertainty in the individual measurements and the uncertainty in the mean value is shown graphically in Fig. 6 for signal y of Figs. 1-3.

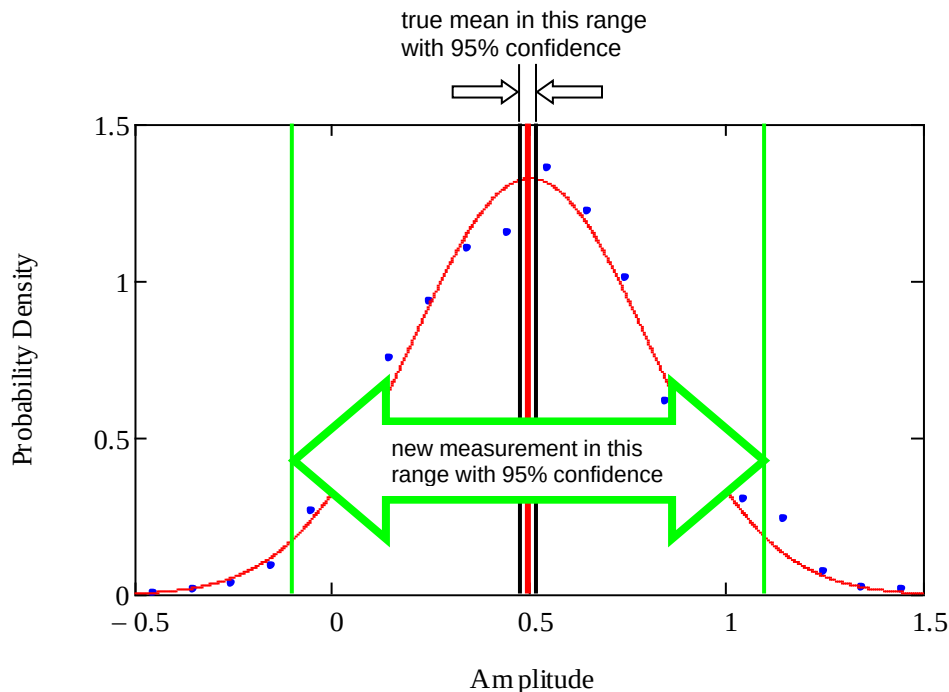


Figure 6: Graphical representation of precision uncertainty for signal y . Notice that the confidence interval on the mean is much smaller than the confidence interval on each measurement. (Recall that signal y comprises 1000 measurements)

Discussion of Confidence Intervals and Probability

Confidence intervals were first described in the 1930s by Neyman [3]. However, there is still ongoing controversy regarding how to describe and interpret them. [4]

The most important thing to keep in mind is that reporting something “with 95% confidence” does **not** mean that there is a 95% probability that true mean is within your estimated confidence interval. The true population mean is (usually) not known, but if it is known, it is either contained within the confidence interval or it is not.

The correct statement is that if you performed the same experiment 100 times, and constructed confidence intervals from your sample measurements using the same procedure each time, you expect that 95 of the confidence intervals you construct would contain the true population mean. However, if you only perform the experiment (sample the population) one time, there is a chance that your results are one of the 5/100 cases where the confidence interval does **not** contain the true

mean, in which case, there is a 0% probability that the mean is contained within your confidence interval.

This distinction can be very important if you are trying to estimate the probability that disastrous outcomes (e.g. a bridge collapsing, a medical treatment causing death) will occur given the results of your measurements and your estimated confidence intervals from a small set of experiments. Matters of safety typically require larger margins of safety, so 95% confidence intervals (representing 1/20 chance of random replication) may need to be elevated to 99.9999% confidence intervals (representing 1/1,000,000 chance of random replication) or higher.

If you are performing multiple independent tests, you may need to select higher confidence level to avoid false positives. [5]

The best approach is to simply give your mean with the confidence interval, and not make claims about the “true” value of the mean. For example, say “The measured pressure for Coke was 362 ± 13 kPa” rather than using the common misconception that “The level of a confidence interval gives the probability that the interval produced by the method employed includes the true value of the parameter”, a misconception that was repeated in online material from a Yale statistics class in the late 1990s. [6]

4.2 Determination of Precision for Small Numbers of Samples

The value of the precision uncertainty of the mean value given in Table 2 must be modified for small numbers of samples ($n < 30$). The error in the derivation of Eq. (7) for small sample sets arises from the fact that it was assumed (in Ref. 1, Sect. 3.6.2) that the true standard deviation of the measurements was equal to the measured standard deviation (σ). For small sample sets, this approximation is no longer valid. (*Do you think you would obtain a good measurement of the standard deviation from only two measurements? Three? Five?*) Instead, the ***t-factor*** is used as a “scaling factor” to increase the computed precision uncertainty for small numbers of measurements. The derivation of the small-sample analog of Eq. (7) is given in Sect. 3.6.3 of Ref. 1, (available in the 2.671 Lab). The 95% confidence interval for the “true mean” μ is

$$\mu_{true} = \mu_{meas} \pm t_{0.025, \nu} \frac{\sigma}{\sqrt{n}}, \quad (95\% \text{ confidence}) \quad (9)$$

where n and σ are defined above, the “degrees of freedom” $\nu \equiv n - 1$, and $t_{0.025, \nu}$ is given in Table 3 below. Degrees of freedom are discussed below the table.

The 95% precision uncertainty in the mean, μ , u_{mean} , is obtained by inspection of Eq. (9) as

$$u_{mean} = t_{0.025, \nu} \frac{\sigma}{\sqrt{n}}. \quad (95\% \text{ confidence}) \quad (10)$$

Table 3 shows that for sample populations of 20 and greater ($\nu \geq 19$) the precision uncertainty from Eq. (10) is just 5% greater than the value calculated for large samples using Eq. (7), for which the *t*-factor is approximately 2.0.

Table 3: Values of the 95% confidence t-distribution¹ ($t_{0.025,\nu}$) for $1 \leq \nu \leq 30$

ν	$t_{0.025}$	ν	$t_{0.025}$	ν	$t_{0.025}$
1	12.706	11	2.201	21	2.080
2	4.303	12	2.179	22	2.074
3	3.182	13	2.160	23	2.069
4	2.776	14	2.145	24	2.064
5	2.571	15	2.131	25	2.060
6	2.447	16	2.120	26	2.056
7	2.365	17	2.110	27	2.052
8	2.306	18	2.101	28	2.048
9	2.262	19	2.093	29	2.045
10	2.228	20	2.086	30	2.042

The preceding subsections described the **statistical** method, which allows determination of the precision uncertainty when multiple measurements of the same parameter are taken. In the next section, we describe the **propagation of uncertainty method**, which allows you to determine precision uncertainty when only one measurement can be taken.

4.3 Degrees of Freedom

In Sect. 4.2 we introduced the parameter ν , defined as $n-1$, where n is the number of points in the data set used to calculate the mean. Identifying ν as $n-1$ is only true for this specific case where a data set is used to determine a single number (in this case, the mean). In fact, ν has a more general meaning – it is the **degrees of freedom** of the system comprising a data set and a parametric relationship [7]. You can consider the degrees of freedom as the number of “redundant” or “extra” data points in the system. Since only one value is required to compute the mean, the remaining $n-1$ points are redundant, and therefore $\nu = n-1$. You will notice from Table 2 that $t_{0.025,\nu}$ is undefined for $\nu = 0$. This follows from the fact that the mean is uniquely determined for a data set of one point. Similarly, suppose you are fitting a line to a data set comprising only two (x,y) pairs. The slope and intercept are uniquely determined, and thus there are no degrees of freedom. Since there are two independent parameters (slope and intercept), the number of redundant points is $n-2$, and thus $\nu = n-2$.

Another way to determine the number of **degrees of freedom** is to recognize that the number of redundant points (or “extra” points) is simply **the number of points, n , minus the number of independent parameters**. If a set of n (x,y) pairs is fit to a quadratic function (as in the case of the trajectory of the ball), there will be 3 adjustable parameters and thus $\nu = n-3$ degrees of freedom.

4.4 Propagation of Uncertainty Method [1] (CASE 2 on the Wiki)

In the Soda Can experiment, only one measurement can be performed of the pressure inside each soda can (a drawback of destructive testing). Since the statistical method is only valid for two or preferably more measurements, the precision uncertainty in your final result (the pressure inside the soda can before opening) cannot be estimated using the statistical method described in the previous section. Instead, it must be estimated from the **propagation of the uncertainties** in each measured quantity used to calculate the pressure. We sometimes refer to this as the “**method**

of partial derivatives” for reasons that will become apparent. This name may make it easier for you to remember how to apply this method.

In many experiments, such as the Soda Can experiment, multiple sources contribute to the uncertainty in the measurement. A method is therefore needed to estimate the total error due to a combination of effects. For a function y of several independent* variables x_i with standard deviations σ_i , a statistical theorem states that the standard deviation of y , σ_y , is [1]

$$y \equiv f(x_1, x_2, \dots, x_n)$$

$$\sigma_y = \sqrt{\left(\frac{\partial f}{\partial x_1} \sigma_1\right)^2 + \left(\frac{\partial f}{\partial x_2} \sigma_2\right)^2 + \dots + \left(\frac{\partial f}{\partial x_n} \sigma_n\right)^2}. \quad (11)$$

However, in many cases the true standard deviations of each independent variable are not known. Instead, if the uncertainties u_i of each independent variable x_i can be estimated, the uncertainty in y , denoted as u_y , can be determined from the uncertainties u_i using

$$u_y = \sqrt{\left(\frac{\partial y}{\partial x_1} u_1\right)^2 + \left(\frac{\partial y}{\partial x_2} u_2\right)^2 + \dots + \left(\frac{\partial y}{\partial x_n} u_n\right)^2} \dots \quad (12)$$

The uncertainties in the independent variables, u_i , may be obtained several ways. The statistical methods of Sect. 4.1 can be used for independent variables that can be measured multiple times (e.g. the diameter of the soda can). For measurements that can only be made once (e.g. voltage when the can is opened), the precision or accuracy of the measuring instrument may be used. Instrument specifications sometimes include both precision and accuracy of the instrument. If the precision is not stated, the accuracy may be used as an upper limit on the precision.

Note that the propagation of uncertainty method (also sometimes referred to as the *partial derivatives method*) can also be used in cases where the desired result (for example the *speed* of sound in a gas) is a simple function of a measured quantity (for example the *slope* of a best-fit line of Time vs Distance). Assume that you have found the uncertainty on the slope of your best-fit line, u_{slope} , using the methods described below. You can then use the following set of equations to find the uncertainty of speed, u_{speed} , given that $speed = 1/slope$.

$$speed = \frac{1}{slope} \quad (13a)$$

$$u_{speed} = \sqrt{\left(\frac{\partial speed}{\partial slope} u_{slope}\right)^2} = \frac{u_{slope}}{slope^2} \quad (13b)$$

Please remember that in such a case, where the variable of interest is the inverse of a measured variable, it is **not true** that the uncertainty of interest is the inverse of the measured uncertainty. You must use the propagation of uncertainty method (with partial derivatives) to find the correct relationship.

This method is only directly applicable when the variables and their uncertainty are independent. If they are covariant, such as coefficients from a least-squares fit, a more detailed implementation is necessary.

4.5 Averaging Values with Uncertainty (CASE 1A on the Wiki)

In many cases you will perform multiple runs of the same experiment. Suppose that each experiment results in a measurement x_i with precision u_i . The precision for each measurement may

have been determined either from a statistical method (resulting from an average as discussed in Sect. 4.1) or from the precision of a parameter obtained from a best-fit of the data (discussed below in Sect. 5) or a propagation of uncertainty calculation (discussed above in Sect. 4.2), or instrument precision. In order to compute the uncertainty in the value obtained from averaging data from the multiple runs together, you need to include both the uncertainty due to averaging the data sets, and the underlying uncertainty of each data set. We describe how to do that in this section, and also on the Wiki (CASE 1A: Averaging Values with Uncertainty).

Assume a set of measurements ($x_1, x_2, \dots x_N$) with uncertainties ($u_1, u_2, \dots u_N$). The uncertainty in the average has two components:

1. Uncertainty due to the averaging of N values together, denoted U_{AVG}
2. Uncertainty due to the uncertainty in each of the underlying N values, denoted U_{INDIV}

The uncertainty due to averaging is computed from the CASE 1 statistical method as:

$$U_{AVG} = \frac{t_{(v=N-1)}\sigma(x_1, x_2, \dots x_N)}{\sqrt{N}}, \quad (14)$$

where σ is the standard deviation of the N measurements and $t_{(v=N-1)}$ is the t-factor for the degrees of freedom, $v = N-1$.

The uncertainty due to the underlying uncertainty of each measurement is computed from the propagation of uncertainty method as follows. Since the average of the x_i values, denoted X_{AVG} is $X_{AVG} = \sum x_i / N$, the method of partial derivatives can be used to find the overall uncertainty due to the uncertainty of the N values as

$$U_{INDIV} = \sqrt{\left(\frac{1}{N}u_1\right)^2 + \left(\frac{1}{N}u_2\right)^2 + \dots + \left(\frac{1}{N}u_N\right)^2}. \quad (15)$$

Since the measurements are assumed to be independent, the **total uncertainty** is therefore

$$U_{TOT} = \sqrt{U_{AVG}^2 + U_{INDIV}^2}. \quad (16)$$

This calculation has assumed that the $\{x_i\}$ are independent with a Gaussian probability distribution.

It is most important to use Case 1A as described in this subsection rather than Case 1 (uncertainty on the mean only, given by Eq. (14)) if the uncertainty in each measurement is larger than or comparable to the spread in the measurements. If the uncertainty in each measurement is less than 10% of the uncertainty in the mean value given by Eq. (14), it should suffice to use Case 1, since the contribution of the individual measurement uncertainty will have a negligible effect on the total uncertainty given in Eq. (16).

5 Reporting Numerical Results

Every numerical result must include 3 components, as shown in Fig. 7: value, uncertainty, and units.

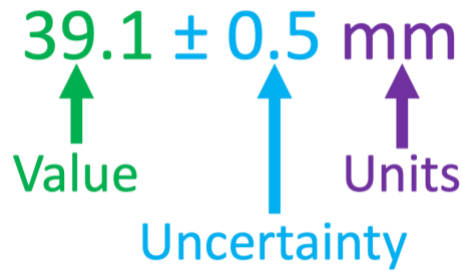


Figure 7: Required elements for reporting numerical results. The number of significant figures for the value and uncertainty are determined by the 2.671 Sig Fig Rules, listed below.

The main point of the sig fig rules is that you must **start with the uncertainty**, write it with **no more than two sig figs** (more would be meaningless information), and then **express the value of the parameter to the same numerical precision** (i.e. the same digit).

The sig fig rules only apply when you are presenting your final numbers; you should never truncate numbers during a calculation as that can cause round-off error, resulting in 10% or more error in the final result.

2.671 Sig Fig Rules

1. Your computed uncertainty should be reported with **at most 2 significant figures**
2. Your value should be expressed to the **same numerical precision** as your uncertainty, i.e. to the same lowest digit

Examples:

$$4.652 \pm 0.053 \text{ m/s}$$

$$(3.45 \pm 0.24) \times 10^{-6} \text{ V}$$

$$(2.455 \pm 0.003) \times 10^{-5} \text{ m/m K}$$

Examples 2 & 3 (scientific notation): value and uncertainty must be expressed to the same power of 10, and enclosed in parentheses as shown. Please format scientific notation as shown, **not** as E-06 or $10^{(-6)}$

Example 3 (1 sig fig on uncertainty): since the uncertainty is so much smaller than the value, only one digit of uncertainty is needed

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